Project Report: Predicting Possible Loan Default with Gradient Boosting Methods

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An advanced machine learning model has been developed to estimate credit risk by predicting the likelihood of loan default. Utilizing diverse datasets, the model evaluates numerous applicant factors such as annual income, credit history, and age through gradient boosting methods. It provides robust, stable predictive results that adapt to shifting consumer behaviors. This model significantly enhances financial institutions' capacity for informed decision-making in lending processes, thereby minimizing financial risks and optimizing risk management strategies.

I. Introduction

Our project is motivated by the 2023 surge in U.S. bank commercial real estate loans, a trend signaling increased future risks and market volatility. This scenario and evolving real estate dynamics could profoundly impact the financial sector. This project was adapted from a Kaggle competition hosted by Home Credit (Kaggle, 2024). The goal of our project, similar to the original competition, is to design a model by applying machine learning techniques to predict how likely a client would default on his loan, and the prediction accuracy and temporal stability evaluate the model. By doing so, we aim to provide companies with a tool to strategize more effectively, helping them mitigate risks and adapt to these emerging uncertainties in the financial landscape.

Accurately predicting loan default probability is crucial for lenders to manage risk effectively. Without reliable predictions, financial institutions could face significant losses from defaulted loans, leading to reduced profitability and potentially harming lending operations. Conversely, institutions that utilize accurate predictive models can optimize interest rates and allocate their capital more efficiently, thereby enhancing profitability.

Moreover, the role of machine learning in credit analysis is also significant, because it introduces advanced techniques to evaluate the risk and decision-making processes. By integrating sophisticated machine learning algorithms, we can process huge amounts of datasets, and uncover hidden patterns and correlations that may be overlooked by traditional statistical methods. The dynamic incorporation of various data sources also facilitates a more nuanced and comprehensive credit risk assessment, thus improving predictive accuracy.

We are convinced that our project will have meaningful real-world applications, potentially influencing financial markets worldwide. By providing a tool that accurately predicts loan default risk, we give lenders a very insightful tool to help them make informed decisions, thereby mitigating financial risk and hence maximizing the profit.

II. Description of Dataset

Our dataset was downloaded from the Kaggle Competition page directly. The dataset is composed of both internal sources and external sources provided by the Home Credit company, which has 1.5 million observations and 465 features, quiet complex. Internal sources come from the actual clients of the company, such as the loan application history of applicants in the past, while external sources are information provided by relative institutions such as credit bureaus and banks. Since all the data came from applicants in

the real-world, the Home Credit masked a substantial number of features in the dataset to prevent any risk of leaking personal information. However, this becomes a headache to us: there are a lot of features that seem to be quite important by observation but we don't know any meaning of it, so it made us realize that we may not use traditional methods based on a very in-depth interpretation of feature meanings. We can conclude that there are two main challenges we faced, one is multiple rows for one observation (case_id), the other is the masked features. From the part of dataset we can see many unknow characters like P55_110_32, P10_68_40…… We don't know their meaning.

case_id	collater_typofvalofguarant_298M	addres_role_871L	contaddr_district_15M	addres_district_368M
6683	P_55_100_32	CONTACT	P10_68_40	P38_92_157
6683	P_55_100_32	PERMANENT	P10_68_40	a55475b1
6683	P_204_92_178	CONTACT	P65_136_169	P7_147_157
6683	P_191_189_75	CONTACT	P10_68_40	P38_92_157
6683	P204_92_178	CONTACT	P164_28_170	P38_92_157

Fig.1 Example of masked features in Dataset

Additionally, one unique feature of this dataset is that each subset in the dataset is associated with a depth number, either 0, 1, or 2. For subsets of depth 0, we can use the information directly to fit the model; however, for subsets of depth 1 or 2, we have to aggregate some information together before we can use it for our models. This is mainly because there is so much less relevant information in subsets of depth 1 or 2. For example, credit bureaus have a full record of applicants' credit history, but it does not help us to make predictions by using someone's credit report from 20 years ago.

Another unique characteristic of the dataset is that some of the features are highly sparse, and when we applied the count() function to find the percentage of sparsity, we shockingly noticed that there are over 40 among 465 features that have sparsity above 85%. It would be another significant problem of the dataset for us to deal with before we proceed with any further analysis.

At the core of this dataset is the "case_id", a unique identifier assigned to each individual's credit profile. This identifier is crucial as it links multiple records from different sources for the same person, reflecting comprehensive financial activities. Besides the "case_id", the dataset contains a few temporal columns like "WEEK_NUM" and "MONTH", providing time-specific insights into the financial behavior of individuals. Thus all the rest columns serve as predictors that describe the financial attributes and behaviors of the individuals. Therefore, our initial phase of Exploratory Data Analysis (EDA) will primarily focus on addressing these data intricacies, ensuring that we thoroughly understand and appropriately prepare the dataset for subsequent in-depth analysis.

III. Project Goal

Our project is focused on developing a cutting-edge model to assess client credit risk regarding loan defaults. Thus in this project, success hinges critically on two pivotal factors: (1) accuracy and (2) stability. Achieving high accuracy ensures that our model reliably predicts client credit risk, minimizing false predictions and enhancing decision-making. Concurrently, stability over time guarantees that the model remains effective despite changing market conditions, ensuring its long-term applicability and value in risk management strategies.

(1) **Accuracy**:

The effectiveness of our model is measured using the Area Under the Receiver Operating Characteristic Curve (AUC), a widely accepted metric for binary classification problems. The AUC metric helps us to understand how well the

model distinguishes between customers (case_id) who will default and those who will not (0/1).

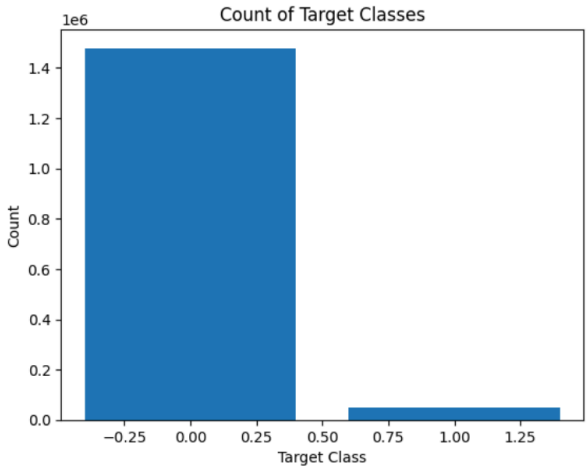


Fig.2 Target Classes for Dataset

For the model to be considered successful and to provide meaningful insights, we aim for an AUC significantly greater than 0.5. An AUC of 0.5 implies that the model's predictions are no better than random chance, akin to flipping a coin.

(2) **Stability**:

To evaluate and quantify the stability of our model, we use the Gini score, a derivative measure from the AUC for each WEEK_NUM. The Gini score provides a snapshot of the model's discriminative power at each weekly interval:

$$gini = 2 \cdot AUC - 1 \quad (1)$$

By examining the Gini score over successive weeks, we can assess the model's consistency in performance.

To comprehensively measure the stability, we employ linear regression to model the relationship between the Gini scores and the weeks:

$$y(gini) = a \cdot x(WEEK_NUM\ AUC) + b \quad (2)$$

Finally we calculate the stability metric:

$$stability\ metric = mean(gini) + 88.0 \cdot min(0, a) - 0.5 \cdot std(residuals) \quad (3)$$

Ensuring stability is essential for the practical deployment of the model, as it must reliably predict outcomes despite the evolving nature of the financial landscape. By actively monitoring and enhancing stability, we can make the model a dependable tool for long-term risk assessment in loan issuance.

In the hypothetical scenario where our model achieves an "ideal" performance, AUC would consistently register at 1 for each weekly evaluation interval (WEEK_NUM). This perfect score translates into a Gini coefficient of 1, indicating absolute discriminative power without any overlap between the default and non-default classes.

IV. Exploratory Data Analysis

We used R for the EDA. As we described above. In our initial phase of data aggregation, we focused on subsets of depth 1 or 2, which presented us with two primary challenges: transformation and distortion of underlying predictors, and the presence of missing values (NAs).

Handling Transformed Predictors: Initially, we considered removing variables obscured by meaningless categorical strings, termed "masking variables." These variables often appear numerical but are disguised for reasons such as privacy. Upon further reflection, we recognized the potential risks of indiscriminately discarding these variables by A. For instance, categorically encoded educational **ASCII** values (e.g., 'A' for university graduates, 'B' for high school graduates) hold significant value despite their opaque labeling. Therefore, we opted to retain such variables provided the categories within them are distinct and unrelated. Conversely, variables with ambiguous relational values, such as obfuscated geographical locations that could influence the analysis due to proximity factors, were excluded.

Addressing Missing Values: We encountered a substantial amount of missing data in our dataset. The straightforward approach of eliminating columns with missing values was deemed inappropriate due to the risk of losing valuable information. Instead, we chose to remove specific instances containing missing values to preserve as

much data integrity as possible. Columns that were severely incomplete or "scant" were entirely removed to maintain dataset quality.

Correlation and Variable Selection: The next step involved determining the correlation between each predictor and the target outcome, specifically defaulting on credit. For numerical predictors, we utilized R's "cor" and "cor.test" functions to assess correlations. The assessment of categorical variables required more nuanced approaches, such as employing Naive Bayes for initial insights. To refine our predictor selection further and ensure robustness in predicting the target, we considered various statistical techniques and machine learning methods, tailoring our approach based on the specific requirements of each model.

After the above step, we removed variables with over 80% missing values, reducing model complexity and avoiding overfitting. Finally, we narrowed the number of features from 465 to 312.

Result for Data Preparation with "depth" =1 or 2

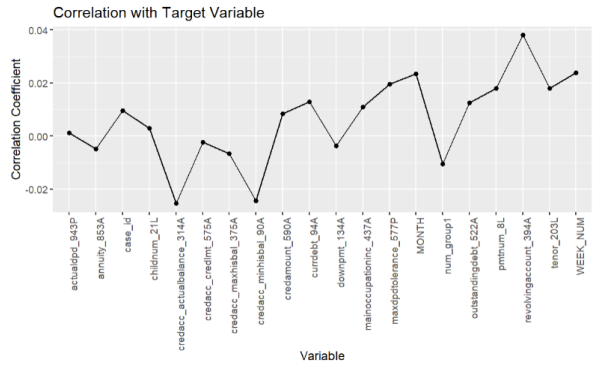


Fig.3 Correlation plot of the data with depth = 1

We can see all the features have a very weak association with the target variable since all the correlation coefficients are from -0.02 to 0.04.

Then we select the column "conts_type_509L" of data with depth = 2 as an example and use this selected feature to show the performance of the data with depth = 2. The result shows it has a very high accuracy of 0.9704 and a sensitivity of 1.0000. Based on its performance and what we have talked about for the initial insights in the Correlation and Variable Selection previously, we chose the Naive Bayes Model, and here is the following result:

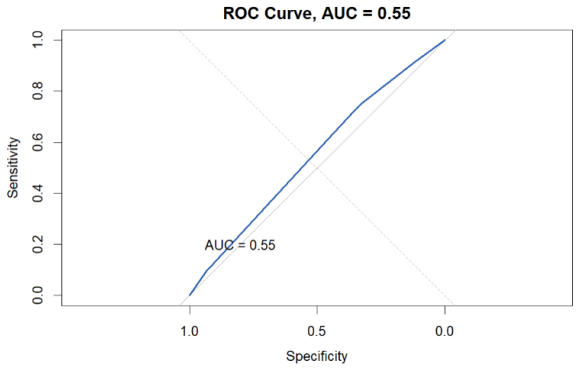


Fig.4 Selected columns of ROC curve with the AUC = 0.5479

One of our goals is for an AUC significantly greater than 0.5, while we can see the AUC by Naive Bayes Model is 0.5479 which is just slightly bigger than 0.5. Thus our initial insight may be not effective, thus we continue to research for more effective methods.

V. Gradient Boosting Methods

In this section, we will show the used model in our project. We mainly used four models to evaluate the accuracy and stability and three of them are based on Gradient Boosting Methods.

Gradient boosting is exceptionally well-suited for our project given its proficiency with complex, heterogeneous datasets. Notably, frameworks like XGBoost and LightGBM are advantageous due to their adaptive loss function optimization, enhancing binary classification tasks. These methods systematically correct previous errors through stage-wise model building, effectively managing non-linear feature-target relationships.

Key strengths include the robust error correction that enhances accuracy by iteratively addressing the mistakes of preceding models. It also handles missing data and overfitting with features like automatic missing value treatment and regularization that prevent overfitting and improve generalizability. This approach is particularly beneficial in scenarios where the relationship between features and the target is complex and non-linear. Efficiently handles data imbalances, which is crucial for applications like loan default predictions, and provides feature importance evaluation. This offers interpretability by

identifying significant predictors, aiding in model refinement and strategic decision-making. These attributes make gradient boosting a powerful tool for maintaining stability and achieving high AUC scores. Overall, the precision, flexibility, and depth of learning offered by gradient boosting make it a compelling choice for achieving high performance in our project's predictive tasks.

A. Mode

The MODE Model (Motivation and Opportunity as Determinants of the Attitude-Behavior Relationship) is an influential framework in social psychology that examines why people's behaviors sometimes align with their attitudes and sometimes do not. This model posits that the strength of the association between an attitude object (e.g., a person, item, or issue) and one's evaluation of it (positive or negative) can significantly influence behavior. According to the MODE model, attitudes that are automatically activated, typically those that are strongly held, are more likely to directly influence behavior.

For our credit risk assessment project, we have chosen to incorporate the MODE model for several compelling reasons such as the direct influence of attitudes on behavior, functional Aspect of attitudes, and adaptability to changes in consumer behavior. Our project aims to refine the prediction of loan default likelihood by not only considering traditional financial metrics but also integrating a psychological perspective that accounts for the complex interplay between attitudes and behaviors in financial decision-making.

The model is trained on 99.8% of the data, with only 0.2% (3000 observations) used for testing.

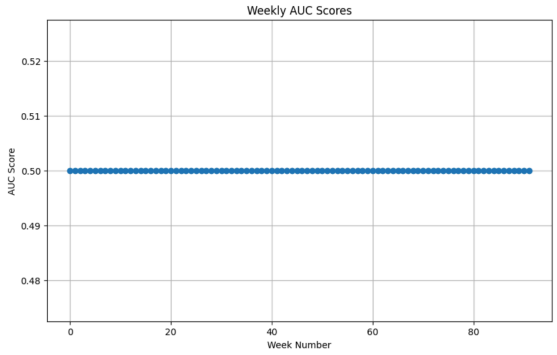


Fig.5 Weekly AUC score for mode model

It achieves an AUC of 0.5, indicating that its performance in distinguishing between default (1) and non-default (0) cases is no better than random chance.

After analyzing the model we find its accuracy is very high up to 0.9686, which is just slightly lower than the accuracy of Naive Bayes (0.9704). From the figure, we can see a high accuracy corresponding with a high false negative rate (same as a low false positive rate), which could be problematic for a loan default prediction model that fails to identify defaults.

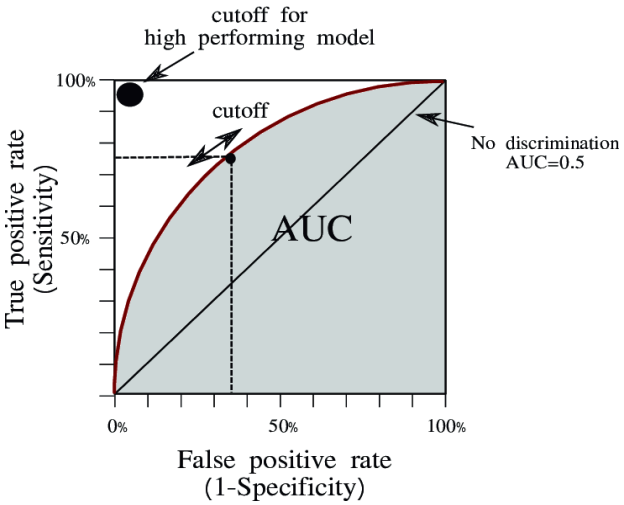


Fig.6 Relationship for AUC with FP Rate

Thus we can conclude that mode plays a very bad performance when predicting the likelihood of loan default, since it may predict every customer would not default.

B. XGBoosting

In our study, we address the challenges of a complex dataset characterized by high sparsity using XGBoost (Extreme Gradient Boosting), which has demonstrated robustness in handling large, incomplete datasets. XGBoost is an advanced implementation of gradient boosting that is not only effective in managing large datasets but also in improving model performance where data is missing.

Model Selection and Configuration: XGBoost stands out for its efficient management of missing data and feature sparsity. It assigns a default

direction for missing observations during tree construction, which is decided by the potential gain in information, thereby maintaining robustness in predictions. The below formula illustrates the gain calculation used in XGBoost, incorporating both first and second-order derivatives of the loss function—specifically, the gradient (g_i) and Hessian (h_i), along with regularization parameters λ (L2 regularization) and γ (minimum loss reduction required to make a further partition on a leaf node of the tree).

$$Gain = \frac{1}{2}\left[\frac{\left(\sum_{i\in L} g_i\right)^2}{\sum_{i\in L} h_i+\lambda} + \frac{\left(\sum_{i\in L} g_i\right)^2}{\sum_{i\in L} h_i+\lambda} - \frac{\left(\sum_{i\in node} g_i\right)^2}{\sum_{i\in node} h_i+\lambda}\right]-\gamma$$

Where:

$$g_i = \partial_{\hat{y}(t-1)}\, l(y_i, \hat{y}^{t-1})$$

and

$$h_i = \partial^2_{\hat{y}(t-1)}\, l(y_i, \hat{y}^{t-1}) \quad (5)$$

Formula 5: This formula integrates gradients and Hessians of the loss function with regularization parameters.

XGBoost enhances the gradient boosting methodology by sequentially building trees where each new tree corrects errors made by the previous ones. This sequential model improvement is guided by minimizing the loss function's derivatives and using the calculated gain to choose optimal splits, as shown in the formula in Figure 1.

Advantages Over Traditional Gradient Boosting: Unlike classical gradient boosting, XGBoost incorporates regularization directly into the model training process to prevent overfitting. This regularization, combined with the algorithm's capability to handle missing values effectively, ensures that XGBoost remains powerful and versatile across various data conditions. The model's ability to automatically handle missing data by assigning them a default direction based on the higher gain reduces the manual pre-processing often required in other models and ensures more accurate outcomes.

Model Validation through Confusion Matrix: The confusion matrix depicted in the figure provides a clear visual of the model's predictive accuracy on the test set. With a low threshold of 0.05, the model achieved remarkably few false negatives (18 out of over 3000 cases), which is crucial for minimizing risk in scenarios like loan approvals where it's preferable to err on the side of caution.

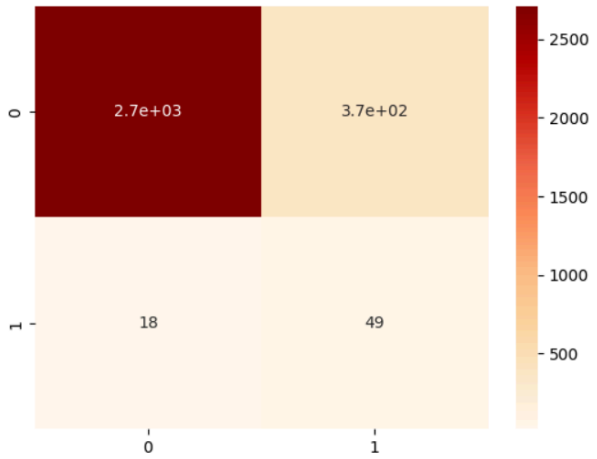


Fig 7: Confusion matrix showing the distribution of true positives, true negatives, false positives, and false negatives.

This matrix confirms the model's robustness, highlighting its capability to correctly classify the vast majority of instances. Such accuracy ensures that risky loans are seldom approved erroneously, supporting effective risk management.

Long-Term Stability Analysis: The stability of the model was further assessed by tracking the Area Under the Receiver Operating Characteristic Curve (AUC) over a period of 91 weeks, as illustrated in Figure 3. The AUC metric serves as a comprehensive measure of model performance, encapsulating both the true positive rate and the false positive rate.

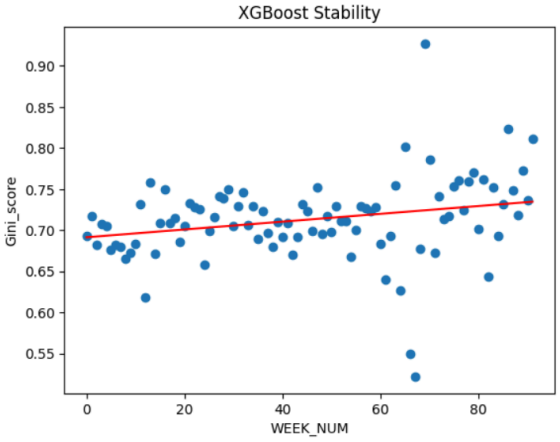


Fig 8: Weekly AUC scores over 91 weeks, showing model performance stability.

As shown, the model not only maintained a consistent AUC score above 0.65 but also exhibited a slight upward trend, indicating an improvement in predictive performance over time. This trend was unexpected and particularly encouraging, suggesting that the model adapts well to new data or possibly benefits from shifts in the underlying data patterns.

Despite the overall positive trends, the presence of outliers in the AUC data points to potential areas of volatility that warrant closer examination. These outliers may arise from anomalous data or external factors affecting specific weeks, underscoring the importance of continuous monitoring and model recalibration to address such discrepancies.

C. LightGBM

Model Overview and Differences: After successfully implementing the XGBoost model, we explored LightGBM, another advanced gradient boosting technique. Both models excel in handling complex datasets, but LightGBM introduces key innovations that optimize performance, particularly for large datasets with high-dimensional features.

LightGBM employs Gradient-based One-Side Sampling (GOSS), which prioritizes data points with the highest gradients for more focused learning on challenging cases. This approach complements its Leaf-wise Tree Growth strategy, where the model grows trees by expanding the most promising leaf, leading to better performance with lower memory usage.

$$Gain = \frac{1}{2}\left[\frac{\left(\sum_{i\in L} g_i\right)^2}{\sum_{i\in L} h_i+\lambda}+\frac{\left(\sum_{i\in L} g_i\right)^2}{\sum_{i\in L} h_i+\lambda}-\frac{\left(\sum_{i\in node} g_i\right)^2}{\sum_{i\in node} h_i+\lambda}\right]-\gamma \quad (6)$$

Formula 6: Same as Formula 5, Gain calculation by gradient boosting

Another notable feature, Exclusive Feature Bundling (EFB), efficiently manages missing data by grouping together features that are rarely active simultaneously, enhancing processing speed and efficiency.

Model Configuration and Evaluation: We evaluated LightGBM's effectiveness through a confusion matrix and an AUC stability plot across 91 weeks. As shown in the confusion matrix, which reveals a high sensitivity of the model, with only 16 false negatives out of over 3000 cases at a decision threshold of 0.05. This low false-negative rate is critical for risk-sensitive applications such as loan approvals.

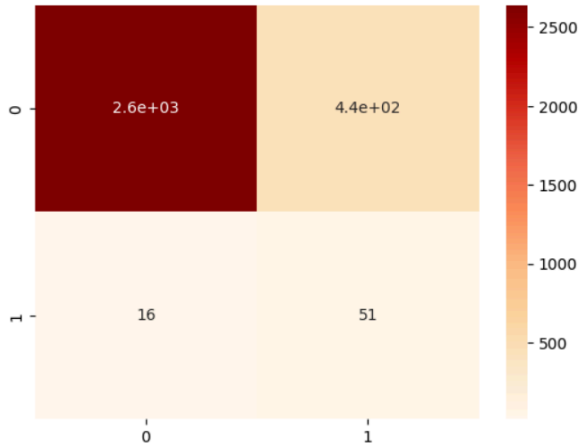


Fig 9: Confusion Matrix of LightGBM, highlighting the low number of false negatives.

AUC Stability Plot: The following figure displays the AUC scores monitored weekly for 91 weeks, illustrating the model's consistent performance.

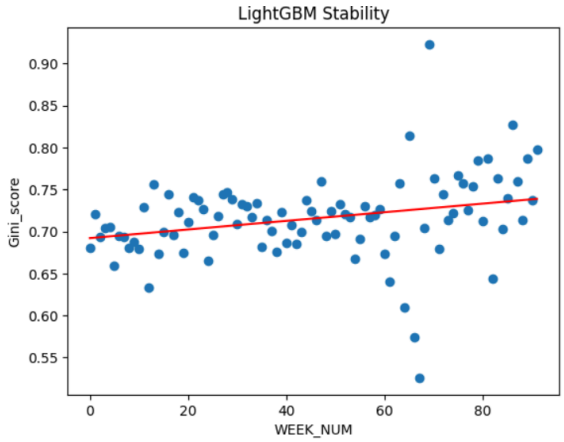


Fig 10: AUC stability plot showing consistent performance of LightGBM over 91 weeks.

The average AUC score remaining above 0.70 indicates a stable and reliable predictive accuracy over time.

D. Stacking

Model 4 employs a stacking approach that utilizes the output probabilities of Light GBM and

XGBoost as input features for a logistic regression model. This method leverages the strength of both ensemble techniques to predict a binary outcome. The logistic regression model is configured with parameters including a seed of 0, a subsample rate of 0.7, a maximum iteration count of 1000, and a depth of 12 for the decision trees.

The prediction function of the logistic regression in the stacking framework is defined as:

$$y = \frac{1}{1 + e^{-\left(\beta_0 + \beta_1 \cdot \frac{0.5 + 0.3}{2}\right)}} \quad (7)$$

Formula 7: Our final predict function with logistic regression

where β_0 and $\beta_1$are the coefficients estimated by the logistic regression, and suppose Light GBM and Xgboost output 0.5, 0.3, which are averaged to form the ensemble input.

Model Performance and Evaluation: The model performance was evaluated using a confusion matrix and the AUC (Area Under the Curve) metric over a period of 91 weeks. The confusion matrix (figure 11) indicated a relatively low number of false negatives (25) at a decision threshold of 0.05, highlighting the model's capability in correctly classifying the positive class. The average AUC over the period (figure 12) was maintained above 0.9, suggesting robust predictive accuracy and stability of the stacking method.

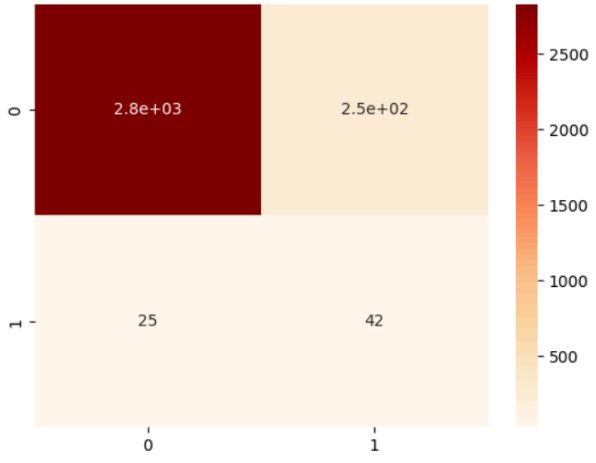


Fig 11: Confusion Matrix of Stacking, highlighting the low number of false negatives.

Further analysis of the stacking method's stability showed consistent Gini scores across the weeks, as depicted in the scatter plot, which does not indicate any significant deviations that could undermine the model's reliability. However, a notable fluctuation in mean weekly AUC was observed around weeks 60 to 80, suggesting variability in the model performance, which did not significantly improve even after applying weighting and data shuffling strategies as shown in the remarks.

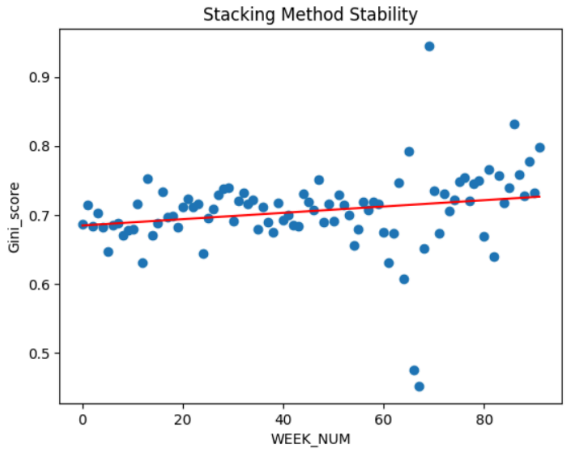


Fig 12: AUC stability plot showing consistent performance of Stacking over 91 weeks.

Model 4 demonstrates a strong potential for high predictive accuracy through the integration of outputs from Light GBM and XGBoost into a logistic regression framework. The consistent performance metrics and stability over time reinforce the model's utility in scenarios requiring reliable predictions. Nonetheless, the observed fluctuations in weekly AUC highlight areas for further investigation and potential refinement in model training or data preprocessing to enhance stability and overall performance.

Remarks for the original data

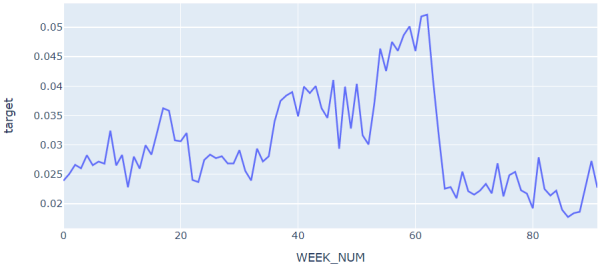


Fig 13: Drastic plunge from the week around 60-70

In our analysis of the model performance over an extended period, we observed fluctuations in the mean week AUC, highlighting particular weeks where the model's prediction accuracy significantly deviated from the trend. As depicted in the line graph (left panel of the figure), there is a pronounced dip around weeks 60 to 80, indicating a potential decrease in model stability or data consistency during this period.

To address these variances and improve model robustness, we experimented with two main strategies: adding weights to the weeks identified with significant deviations and shuffling the dataset to remodel the training process which aimed at enhancing the model's ability to generalize across different temporal segments of the dataset, thus potentially stabilizing the week-to-week performance.

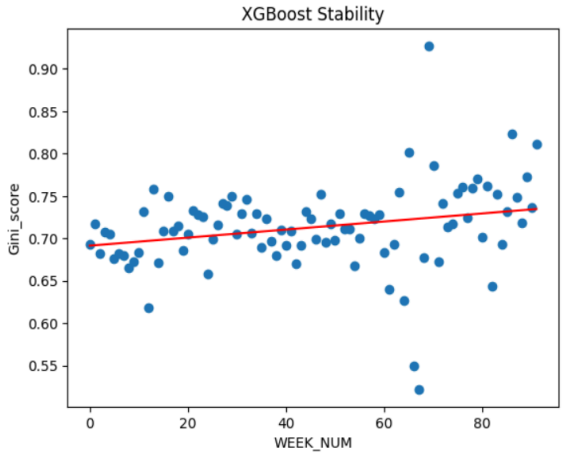


Fig 14: Stability after adjustment

From the above figure shows, we find that despite these adjustments, the scatter plot on the right panel, which displays the Gini score across weeks, suggests that there was no significant improvement in the model's stability. The trend line across the weeks shows a mild positive slope but lacks a substantial shift in variability or overall performance enhancement. This outcome indicates that while the model generally maintains a moderate level of prediction accuracy, as evidenced by the Gini scores ranging from approximately 0.65 to 0.85, the adjustments made did not markedly influence the model's performance across the observed weeks.

This analysis underscores the challenges in achieving consistent model performance through temporal adjustments and highlights the need for further exploration into more sophisticated methods or additional features that could contribute to a more stable prediction over time.

VI. Conclusion

This study has extensively analyzed the performance of advanced machine learning models in the estimation of credit risk, focusing particularly on the implications of model accuracy and error types on lender profitability and risk management. The models evaluated include LightGBM, XGBoost, and a Stacking model, each possessing unique features and capabilities.

Our results, as shown in the comparative analysis, indicate that LightGBM outperformed the other models in terms of stability and AUC (Area Under the Curve) metrics. LightGBM recorded an AUC of 0.860525 and demonstrated fewer false negatives (FN=18) compared to XGBoost (AUC=0.859546, FN=16) and the Stacking model (AUC=0.856855, FN=25). The stability metric further solidified LightGBM's superiority with a score of 0.697172, compared to XGBoost's 0.694776 and the Stacking model's 0.685807.

The significance of these results lies in the impact of Type II errors (False Negatives) and Type I errors (False Positives) on the operational dynamics of financial institutions. False Negative errors pose a significant risk as they involve estimating an applicant as creditworthy when they are likely to default, directly affecting the losses and provisions of the business. Conversely, False Positive errors, where applicants are incorrectly deemed likely to default, affect the profitability by potentially denying credit to viable customers.

Given these considerations, LightGBM's superior performance in minimizing False Negatives, while maintaining competitive accuracy as evidenced by its AUC score, suggests that it offers the best balance between risk management and profitability. This model efficiently handles the trade-offs between denying loans to potential defaulters and approving loans to reliable borrowers, which is critical for minimizing credit losses while maximizing financial returns.

In conclusion, the application of LightGBM in credit risk assessment provides a robust tool for financial institutions aiming to enhance their decision-making processes. Its ability to effectively

manage the trade-offs between different types of predictive errors makes it an invaluable asset in the challenging landscape of credit risk management. Future research should consider exploring the integration of these models with real-time data processing to dynamically adapt to new trends in borrower behavior and economic conditions. This approach could further refine the predictions and enhance the financial sustainability of lending operations.

VII. Limation & Future Direction

The complexities of our predictive model stem from the intricate relationships among variables, making intuitive explanations of their interactions with loan defaults challenging. Additionally, the task of pruning predictors is particularly daunting given the vast permutations (2^{312}) and limited understanding of some variables' inherent meanings.

A notable concern is a pronounced dip in model performance during weeks 60-70, a pattern that persisted despite efforts like data shuffling and weighting, indicating potential underlying issues not addressed by these methods. Furthermore, our approach to handling string variables through the ASCII value comparisons, although not ideal, remains our current method due to the lack of a more sophisticated aggregation strategy.

To enhance the robustness and accuracy of our model, future work should focus on developing more refined techniques for feature aggregation, especially for string variables. Improving our understanding of each predictor's role and effect could lead to more effective feature selection and potentially resolve issues observed in specific timeframes, such as the performance drop in weeks 60-70. Adopting advanced machine learning techniques that can automatically detect and adjust to non-linear relationships and interactions among features may also prove beneficial. This progression aims to not only advance the model's performance but also provide clearer insights into the complex dynamics of loan default predictions.

VIII. Appendix

Kaggle. (2024). Home Credit Default Risk Data. Retrieved from https://www.kaggle.com/competitions/home-credit-credit-risk-model-stability/data